Name: Gaurav Joshi **Batch: T13** **Roll No:50**

EXPERIMENT 9

Aim:
To understand Docker Architecture and Container Life Cycle, install Docker and execute docker commands to manage images and interact with containers.

Theory:
Docker is a popular platform that enables developers to build, package, and deploy applications as lightweight, portable, and self-sufficient containers. These containers encapsulate all the necessary dependencies and libraries required for an application to run, ensuring consistency across different environments. Here is a theoretical overview of Docker:

1)Containerization:
Docker utilizes containerization technology to create isolated environments for applications. Containers are lightweight, standalone, and executable packages that include everything needed to run an application, such as code, runtime, system tools, libraries, and settings. This isolation ensures that applications run consistently across different environments, from development to production.

2)Docker Engine:
At the core of Docker is the Docker Engine, which is responsible for building, running, and managing containers. It consists of the Docker daemon, which manages containers, images, networks, and volumes, and the Docker client, which allows users to interact with the daemon through the Docker API.

3)Docker Images:
Docker images are read-only templates used to create containers. They contain the application code, runtime, libraries, dependencies, and other files needed to run the application. Images are built using Dockerfiles, which are text files that define the steps needed to create the image.

4)Docker Containers:
Containers are instances of Docker images that are running as isolated processes on a host machine. They are lightweight, portable, and can be easily started, stopped, moved, and deleted. Containers provide a consistent environment for applications to run, regardless of the underlying infrastructure.

Benefits of Docker:

1)Portability: Docker containers can run on any platform that supports Docker, making it easy to deploy applications across different environments.

2)Efficiency: Containers share the host OS kernel, reducing overhead and improving resource utilization.

3)Isolation: Containers provide a level of isolation that helps prevent conflicts between applications and dependencies.

4)Scalability: Docker enables easy scaling of applications by quickly spinning up additional Containers.

5)Consistency: Docker ensures that applications run the same way in development, testing, and production environments.

Output:

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```
C:\Users\L15\Desktop\sepm_t13_49>docker rm nifty_stonebraker
nifty_stonebraker

C:\Users\L15\Desktop\sepm_t13_49>docker images
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/welcome-to-docker:latest      c4d56c24da4f       22.2MB         6.03MB
hello-world:latest                   ef54e839ef54       25.9kB         9.52kB    U
redis:latest                         009cc37796fb        204MB         55.3MB

C:\Users\L15\Desktop\sepm_t13_49>docker run -d redis
b7a18b45f2173a4e237a702f46d1583afd43be896a441cd01f0635dd2e63119e

C:\Users\L15\Desktop\sepm_t13_49>docker exec -d b7a18b45f2173a4e237a702f46d1583afd43be896a441cd01f0635dd2e63119e touch/tmp/execWorks
Error response from daemon: OCI runtime exec failed: exec failed: unable to start container process: exec: "touch/tmp/execWorks": stat touch/tmp/execWorks: no such file or directory

C:\Users\L15\Desktop\sepm_t13_49>docker exec -d b7a18b45f2173a4e237a702f46d1583afd43be896a441cd01f0635dd2e63119e touch /tmp/execWorks

C:\Users\L15\Desktop\sepm_t13_49>docker exec -it b7a18b45f2173a4e237a702f46d1583afd43be896a441cd01f0635dd2e63119e bash
root@b7a18b45f217:/data# docker exec -d b7a18b45f2173a4e237a702f46d1583afd43be896a441cd01f0635dd2e63119e touch/tmp/execWorks
bash: docker: command not found
root@b7a18b45f217:/data# ls/tmp
bash: ls/tmp: No such file or directory
root@b7a18b45f217:/data# ls /temp
ls: cannot access '/temp': No such file or directory
root@b7a18b45f217:/data# ls /tmp
execWorks  rustix_test_can_compile
root@b7a18b45f217:/data# exit
exit

C:\Users\L15\Desktop\sepm_t13_49>docker commit b7a18b45f2173a4e237a702f46d1583afd43be896a441cd01f0635dd2e63119e myredis:version1
sha256:9199aba240ab82112cca962eff1dea4a146ea0a5956aff3d70590554cf4545f9

C:\Users\L15\Desktop\sepm_t13_49>docker images
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/welcome-to-docker:latest      c4d56c24da4f       22.2MB         6.03MB
hello-world:latest                   ef54e839ef54       25.9kB         9.52kB    U
myredis:version1                     9199aba240ab        202MB         53.2MB
redis:latest                         009cc37796fb        204MB         55.3MB    U

C:\Users\L15\Desktop\sepm_t13_49>docker run redis
Starting Redis Server
1:C 30 Mar 2026 08:21:43.310 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 30 Mar 2026 08:21:43.310 * Redis version=8.6.2, bits=64, commit=00000000, modified=1, pid=1, just started
1:C 30 Mar 2026 08:21:43.310 * Configuration loaded
1:M 30 Mar 2026 08:21:43.311 * monotonic clock: POSIX clock_gettime
1:M 30 Mar 2026 08:21:43.311 * Running mode=standalone, port=6379.
1:M 30 Mar 2026 08:21:43.311 * <bf> RedisBloom version 8.6.0 (Git=unknown)
1:M 30 Mar 2026 08:21:43.311 * <bf> Registering configuration options: [
1:M 30 Mar 2026 08:21:43.311 * <bf>     { bf-error-rate       :      0.01 }
1:M 30 Mar 2026 08:21:43.311 * <bf>     { bf-initial-size     :       100 }
1:M 30 Mar 2026 08:21:43.311 * <bf>     { bf-expansion-factor :         2 }
1:M 30 Mar 2026 08:21:43.311 * <bf>     { cf-bucket-size      :         2 }
1:M 30 Mar 2026 08:21:43.311 * <bf>     { cf-initial-size     :      1024 }
1:M 30 Mar 2026 08:21:43.311 * <bf>     { cf-max-iterations   :        20 }
```

```
                    "IPAddress": "172.17.0.2",
                    "MacAddress": "be:e0:71:5e:0d:33",
                    "IPPrefixLen": 16,
                    "IPv6Gateway": "",
                    "GlobalIPv6Address": "",
                    "GlobalIPv6PrefixLen": 0,
                    "DNSNames": null
                }
            }
        },
        "ImageManifestDescriptor": {
            "mediaType": "application/vnd.oci.image.manifest.v1+json",
            "digest": "sha256:17a235ee842cd21d8bcb7f11a49fb177cbeaf5537996304a0089e94412de2ed5",
            "size": 2286,
            "annotations": {
                "com.docker.official-images.bashbrew.arch": "amd64",
                "org.opencontainers.image.base.digest": "sha256:edc9450a9fe37d30b508808052f8d0564e3ed9eaf565e043c6f5632957f7381e",
                "org.opencontainers.image.base.name": "debian:trixie-slim",
                "org.opencontainers.image.created": "2026-03-24T18:12:25Z",
                "org.opencontainers.image.revision": "8c81e2a44cae9258294718d767ce594e5cbbf20e",
                "org.opencontainers.image.source": "https://github.com/redis/docker-library-redis.git#8c81e2a44cae9258294718d767ce594e5cbbf20e:debian",
                "org.opencontainers.image.url": "https://hub.docker.com/_/redis",
                "org.opencontainers.image.version": "8.6.2"
            },
            "platform": {
                "architecture": "amd64",
                "os": "linux"
            }
        }
    }
]

C:\Users\L15\Desktop\sepm_t13_49>docker restart nifty_stonebraker
nifty_stonebraker

C:\Users\L15\Desktop\sepm_t13_49>docker stop nifty_stonebraker
nifty_stonebraker

C:\Users\L15\Desktop\sepm_t13_49>docker ps-a
docker: unknown command: docker ps-a

Run 'docker --help' for more information

C:\Users\L15\Desktop\sepm_t13_49>docker ps -a
CONTAINER ID   IMAGE         COMMAND                  CREATED                 STATUS                      PORTS     NAMES
2ced95d05ddb   redis         "docker-entrypoint.s…"   Less than a second ago   Exited (0) 22 seconds ago             nifty_stonebraker
222596f84fc0   hello-world   "/hello"                 3 weeks ago             Exited (0) 3 weeks ago                gracious_wilson

C:\Users\L15\Desktop\sepm_t13_49>docker rm nifty_stonebraker
nifty_stonebraker

C:\Users\L15\Desktop\sepm_t13_49>docker images
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/welcome-to-docker:latest      c4d56c24da4f       22.2MB         6.03MB
hello-world:latest                   ef54e839ef54       25.9kB         9.52kB    U
```

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```
        },
        "Mounts": [],
        "Config": {
            "Hostname": "2ced95d05ddb",
            "Domainname": "",
            "User": "",
            "AttachStdin": false,
            "AttachStdout": true,
            "AttachStderr": true,
            "ExposedPorts": {
                "6379/tcp": {}
            },
            "Tty": false,
            "OpenStdin": false,
            "StdinOnce": false,
            "Env": [
                "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
            ],
            "Cmd": [
                "redis-server"
            ],
            "Image": "redis",
            "Volumes": null,
            "WorkingDir": "/data",
            "Entrypoint": [
                "docker-entrypoint.sh"
            ],
            "Labels": {},
            "StopTimeout": 1
        },
        "NetworkSettings": {
            "SandboxID": "dbb39db83e888472fcac73d62b00004d4ae67036df962de0c71c378376af5dc5",
            "SandboxKey": "/var/run/docker/netns/dbb39db83e88",
            "Ports": {
                "6379/tcp": null
            },
            "Networks": {
                "bridge": {
                    "IPAMConfig": null,
                    "Links": null,
                    "Aliases": null,
                    "DriverOpts": null,
                    "GwPriority": 0,
                    "NetworkID": "24b80ae507945e740f638cb849634e006550829c00e7fb34177560d70dbe7a8b",
                    "EndpointID": "9d3a90a743c0d91277b17fba8331659cbd35a26f4447f20fa39109d0e8d2a03f",
                    "Gateway": "172.17.0.1",
                    "IPAddress": "172.17.0.2",
                    "MacAddress": "be:e0:71:5e:0d:33",
                    "IPPrefixLen": 16,
                    "IPv6Gateway": "",
                    "GlobalIPv6Address": "",
                    "GlobalIPv6PrefixLen": 0,
                    "DNSNames": null
                }
            }
        },
```

```
            "DnsOptions": [],
            "DnsSearch": [],
            "ExtraHosts": null,
            "GroupAdd": null,
            "IpcMode": "private",
            "Cgroup": "",
            "Links": null,
            "OomScoreAdj": 0,
            "PidMode": "",
            "Privileged": false,
            "PublishAllPorts": false,
            "ReadonlyRootfs": false,
            "SecurityOpt": null,
            "UTSMode": "",
            "UsernsMode": "",
            "ShmSize": 67108864,
            "Runtime": "runc",
            "Isolation": "",
            "CpuShares": 0,
            "Memory": 0,
            "NanoCpus": 0,
            "CgroupParent": "",
            "BlkioWeight": 0,
            "BlkioWeightDevice": [],
            "BlkioDeviceReadBps": [],
            "BlkioDeviceWriteBps": [],
            "BlkioDeviceReadIOps": [],
            "BlkioDeviceWriteIOps": [],
            "CpuPeriod": 0,
            "CpuQuota": 0,
            "CpuRealtimePeriod": 0,
            "CpuRealtimeRuntime": 0,
            "CpusetCpus": "",
            "CpusetMems": "",
            "Devices": [],
            "DeviceCgroupRules": null,
            "DeviceRequests": null,
            "MemoryReservation": 0,
            "MemorySwap": 0,
            "MemorySwappiness": null,
            "OomKillDisable": null,
            "PidsLimit": null,
            "Ulimits": [],
            "CpuCount": 0,
            "CpuPercent": 0,
            "IOMaximumIOps": 0,
            "IOMaximumBandwidth": 0,
            "MaskedPaths": [
                "/proc/acpi",
                "/proc/asound",
                "/proc/interrupts",
                "/proc/kcore",
                "/proc/keys",
                "/proc/latency_stats",
                "/proc/sched_debug",
                "/proc/scsi",
```

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```
exit

C:\Users\L15\Desktop\sepm_t13_49>docker inspect nifty_stonebraker
[
    {
        "Id": "2ced95d05ddb00fa7e23152524a88bad14839685722624f6fec489d0e7be88e6",
        "Created": "2026-03-30T07:57:25.533107503Z",
        "Path": "docker-entrypoint.sh",
        "Args": [
            "redis-server"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 589,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-03-30T08:06:35.396478152Z",
            "FinishedAt": "2026-03-30T08:02:35.427597329Z"
        },
        "Image": "sha256:009cc37796fbdbe1b631b4cc0582bed167e5e403ed8bcd06f77eb6cb5aeb6f93",
        "ResolvConfPath": "/var/lib/docker/containers/2ced95d05ddb00fa7e23152524a88bad14839685722624f6fec489d0e7be88e6/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/2ced95d05ddb00fa7e23152524a88bad14839685722624f6fec489d0e7be88e6/hostname",
        "HostsPath": "/var/lib/docker/containers/2ced95d05ddb00fa7e23152524a88bad14839685722624f6fec489d0e7be88e6/hosts",
        "LogPath": "/var/lib/docker/containers/2ced95d05ddb00fa7e23152524a88bad14839685722624f6fec489d0e7be88e6/2ced95d05ddb00fa7e23152524a88bad14839685722624f6fec489d0e7be88e6-json.log",
        "Name": "/nifty_stonebraker",
        "RestartCount": 0,
        "Driver": "overlayfs",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
            },
            "NetworkMode": "bridge",
            "PortBindings": {},
            "RestartPolicy": {
                "Name": "no",
                "MaximumRetryCount": 0
            },
            "AutoRemove": false,
            "VolumeDriver": "",
            "VolumesFrom": null,
            "ConsoleSize": [
                30,
                120
```

```
1:M 30 Mar 2026 08:06:35.540 * <ReJSON> version: 80600 git sha: unknown branch: unknown
1:M 30 Mar 2026 08:06:35.540 * <ReJSON> Exported RedisJSON_V1 API
1:M 30 Mar 2026 08:06:35.540 * <ReJSON> Exported RedisJSON_V2 API
1:M 30 Mar 2026 08:06:35.540 * <ReJSON> Exported RedisJSON_V3 API
1:M 30 Mar 2026 08:06:35.540 * <ReJSON> Exported RedisJSON_V4 API
1:M 30 Mar 2026 08:06:35.540 * <ReJSON> Exported RedisJSON_V5 API
1:M 30 Mar 2026 08:06:35.540 * <ReJSON> Exported RedisJSON_V6 API
1:M 30 Mar 2026 08:06:35.540 * <ReJSON> Enabled diskless replication
1:M 30 Mar 2026 08:06:35.540 * <ReJSON> Initialized shared string cache, thread safe: true.
1:M 30 Mar 2026 08:06:35.540 * Module 'ReJSON' loaded from /usr/local/lib/redis/modules//rejson.so
1:M 30 Mar 2026 08:06:35.540 * <search> Acquired RedisJSON_V6 API
1:M 30 Mar 2026 08:06:35.541 * Server initialized
1:M 30 Mar 2026 08:06:35.541 * <search> Changing workers threadpool size from 0 to 4
1:M 30 Mar 2026 08:06:35.541 * <search> Enabled workers threadpool of size 4
1:M 30 Mar 2026 08:06:35.541 * <search> Loading event started
1:M 30 Mar 2026 08:06:35.541 * Loading RDB produced by version 8.6.2
1:M 30 Mar 2026 08:06:35.541 * RDB age 240 seconds
1:M 30 Mar 2026 08:06:35.541 * RDB memory usage when created 1.26 Mb
1:M 30 Mar 2026 08:06:35.541 * Done loading RDB, keys loaded: 0, keys expired: 0.
1:M 30 Mar 2026 08:06:35.541 * <search> Changing workers threadpool size from 4 to 0
1:M 30 Mar 2026 08:06:35.541 * <search> Scheduling config_reduce_threads_job to remove all 4 threads when empty
1:M 30 Mar 2026 08:06:35.541 * <search> Disabled workers threadpool of size 4
1:M 30 Mar 2026 08:06:35.541 * <search> Loading event ended successfully
1:M 30 Mar 2026 08:06:35.541 * DB loaded from disk: 0.000 seconds
1:M 30 Mar 2026 08:06:35.541 * Ready to accept connections tcp
1:M 30 Mar 2026 08:06:35.541 # WARNING: Redis does not require authentication and is not protected by network restrictions. Redis will accept connections from any IP address on any network interface.

C:\Users\L15\Desktop\sepm_t13_49>docker exec -it nifty_stonebraker bash
root@2ced95d05ddb:/data# redis-cli
127.0.0.1:6379> PING
PONG
127.0.0.1:6379> SET name "Docker"
OK
127.0.0.1:6379> GET name
"Docker"
127.0.0.1:6379> exit
root@2ced95d05ddb:/data# exit
exit

C:\Users\L15\Desktop\sepm_t13_49>docker inspect nifty_stonebraker
[
    {
        "Id": "2ced95d05ddb00fa7e23152524a88bad14839685722624f6fec489d0e7be88e6",
        "Created": "2026-03-30T07:57:25.533107503Z",
        "Path": "docker-entrypoint.sh",
        "Args": [
            "redis-server"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
```

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```
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Initialized shared string cache, thread safe: true.
1:M 30 Mar 2026 07:57:25.851 * Module 'ReJSON' loaded from /usr/local/lib/redis/modules//rejson.so
1:M 30 Mar 2026 07:57:25.851 * <search> Acquired RedisJSON_V6 API
1:M 30 Mar 2026 07:57:25.851 * Server initialized
1:M 30 Mar 2026 07:57:25.851 * Ready to accept connections tcp
1:M 30 Mar 2026 07:57:25.851 # WARNING: Redis does not require authentication and is not protected by network restrictions. Redis will accept connections from any IP address on any network interface.
1:signal-handler (1774857755) Received SIGINT scheduling shutdown...
1:M 30 Mar 2026 08:02:35.407 * User requested shutdown...
1:M 30 Mar 2026 08:02:35.407 * Saving the final RDB snapshot before exiting.
1:M 30 Mar 2026 08:02:35.417 * BGSAVE done, 0 keys saved, 0 keys skipped, 88 bytes written.
1:M 30 Mar 2026 08:02:35.422 * DB saved on disk
1:M 30 Mar 2026 08:02:35.422 # Redis is now ready to exit, bye bye...
Starting Redis Server
1:C 30 Mar 2026 08:06:35.534 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 30 Mar 2026 08:06:35.534 * Redis version=8.6.2, bits=64, commit=00000000, modified=1, pid=1, just started
1:C 30 Mar 2026 08:06:35.534 * Configuration loaded
1:M 30 Mar 2026 08:06:35.535 * monotonic clock: POSIX clock_gettime
1:M 30 Mar 2026 08:06:35.535 * Running mode=standalone, port=6379.
1:M 30 Mar 2026 08:06:35.536 * <bf> RedisBloom version 8.6.0 (Git=unknown)
1:M 30 Mar 2026 08:06:35.536 * <bf> Registering configuration options: [
1:M 30 Mar 2026 08:06:35.536 * <bf>     { bf-error-rate       :      0.01 }
1:M 30 Mar 2026 08:06:35.536 * <bf>     { bf-initial-size     :       100 }
1:M 30 Mar 2026 08:06:35.536 * <bf>     { bf-expansion-factor :         2 }
1:M 30 Mar 2026 08:06:35.536 * <bf>     { cf-bucket-size      :         2 }
1:M 30 Mar 2026 08:06:35.536 * <bf>     { cf-initial-size     :      1024 }
1:M 30 Mar 2026 08:06:35.536 * <bf>     { cf-max-iterations   :        20 }
1:M 30 Mar 2026 08:06:35.536 * <bf>     { cf-expansion-factor :         1 }
1:M 30 Mar 2026 08:06:35.536 * <bf>     { cf-max-expansions   :        32 }
1:M 30 Mar 2026 08:06:35.536 * <bf> ]
1:M 30 Mar 2026 08:06:35.536 * Module 'bf' loaded from /usr/local/lib/redis/modules//redisbloom.so
1:M 30 Mar 2026 08:06:35.538 * <search> Redis version found by RedisSearch : 8.6.2 - oss
1:M 30 Mar 2026 08:06:35.538 * <search> RediSearch version 8.6.0 (Git=7782b97)
1:M 30 Mar 2026 08:06:35.538 * <search> Low level api version 1 initialized successfully
1:M 30 Mar 2026 08:06:35.538 * <search> gc: ON, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200, query timeout (ms): 500, timeout policy: return, oom policy: return, cursor read si
ze: 1000, cursor max idle (ms): 300000, max doctable size: 1000000, max number of search results:  1000000, default scorer: BM25STD,
1:M 30 Mar 2026 08:06:35.538 * <search> Initialized thread pools!
1:M 30 Mar 2026 08:06:35.538 * <search> Disabled workers threadpool of size 0
1:M 30 Mar 2026 08:06:35.538 * <search> Subscribe to config changes
1:M 30 Mar 2026 08:06:35.538 * <search> Subscribe to cluster slot migration events
1:M 30 Mar 2026 08:06:35.538 * <search> Enabled role change notification
1:M 30 Mar 2026 08:06:35.538 * <search> Cluster configuration: AUTO partitions, type: 0, coordinator timeout: 0ms
1:M 30 Mar 2026 08:06:35.539 * Module 'search' loaded from /usr/local/lib/redis/modules//redisearch.so
1:M 30 Mar 2026 08:06:35.539 * <timeseries> RedisTimeSeries version 80600, git_sha=05fd355db748676861dc4c17d19c8c1ca74c0154
1:M 30 Mar 2026 08:06:35.539 * <timeseries> Redis version found by RedisTimeSeries : 8.6.2 - oss
1:M 30 Mar 2026 08:06:35.539 * <timeseries> Registering configuration options: [
1:M 30 Mar 2026 08:06:35.539 * <timeseries>     { ts-compaction-policy   :             }
1:M 30 Mar 2026 08:06:35.539 * <timeseries>     { ts-num-threads         :           3 }
1:M 30 Mar 2026 08:06:35.539 * <timeseries>     { ts-retention-policy    :           0 }
1:M 30 Mar 2026 08:06:35.539 * <timeseries>     { ts-duplicate-policy    :       block }
1:M 30 Mar 2026 08:06:35.539 * <timeseries>     { ts-chunk-size-bytes    :        4096 }
1:M 30 Mar 2026 08:06:35.539 * <timeseries>     { ts-encoding            :  compressed }
1:M 30 Mar 2026 08:06:35.539 * <timeseries>     { ts-ignore-max-time-diff:           0 }
1:M 30 Mar 2026 08:06:35.539 * <timeseries>     { ts-ignore-max-val-diff :    0.000000 }
1:M 30 Mar 2026 08:06:35.539 * <timeseries> ]
1:M 30 Mar 2026 08:06:35.539 * <timeseries> Detected redis oss
1:M 30 Mar 2026 08:06:35.539 * <timeseries> Subscribe to ASM events
```

```
C:\Users\L15\Desktop\sepm_t13_49>docker logs nifty_stonebraker
Starting Redis Server
1:C 30 Mar 2026 07:57:25.847 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 30 Mar 2026 07:57:25.847 * Redis version=8.6.2, bits=64, commit=00000000, modified=1, pid=1, just started
1:C 30 Mar 2026 07:57:25.847 * Configuration loaded
1:M 30 Mar 2026 07:57:25.848 * monotonic clock: POSIX clock_gettime
1:M 30 Mar 2026 07:57:25.848 * Running mode=standalone, port=6379.
1:M 30 Mar 2026 07:57:25.848 * <bf> RedisBloom version 8.6.0 (Git=unknown)
1:M 30 Mar 2026 07:57:25.848 * <bf> Registering configuration options: [
1:M 30 Mar 2026 07:57:25.848 * <bf>     { bf-error-rate       :      0.01 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { bf-initial-size     :       100 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { bf-expansion-factor :         2 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-bucket-size      :         2 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-initial-size     :      1024 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-max-iterations   :        20 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-expansion-factor :         1 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-max-expansions   :        32 }
1:M 30 Mar 2026 07:57:25.848 * <bf> ]
1:M 30 Mar 2026 07:57:25.849 * Module 'bf' loaded from /usr/local/lib/redis/modules//redisbloom.so
1:M 30 Mar 2026 07:57:25.850 * <search> Redis version found by RedisSearch : 8.6.2 - oss
1:M 30 Mar 2026 07:57:25.850 * <search> RediSearch version 8.6.0 (Git=7782b97)
1:M 30 Mar 2026 07:57:25.850 * <search> Low level api version 1 initialized successfully
1:M 30 Mar 2026 07:57:25.850 * <search> gc: ON, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200, query timeout (ms): 500, timeout policy: return, oom policy: return, cursor read si
ze: 1000, cursor max idle (ms): 300000, max doctable size: 1000000, max number of search results:  1000000, default scorer: BM25STD,
1:M 30 Mar 2026 07:57:25.850 * <search> Initialized thread pools!
1:M 30 Mar 2026 07:57:25.850 * <search> Disabled workers threadpool of size 0
1:M 30 Mar 2026 07:57:25.850 * <search> Subscribe to config changes
1:M 30 Mar 2026 07:57:25.850 * <search> Subscribe to cluster slot migration events
1:M 30 Mar 2026 07:57:25.850 * <search> Enabled role change notification
1:M 30 Mar 2026 07:57:25.850 * <search> Cluster configuration: AUTO partitions, type: 0, coordinator timeout: 0ms
1:M 30 Mar 2026 07:57:25.850 * Module 'search' loaded from /usr/local/lib/redis/modules//redisearch.so
1:M 30 Mar 2026 07:57:25.850 * <timeseries> RedisTimeSeries version 80600, git_sha=05fd355db748676861dc4c17d19c8c1ca74c0154
1:M 30 Mar 2026 07:57:25.851 * <timeseries> Redis version found by RedisTimeSeries : 8.6.2 - oss
1:M 30 Mar 2026 07:57:25.851 * <timeseries> Registering configuration options: [
1:M 30 Mar 2026 07:57:25.851 * <timeseries>     { ts-compaction-policy   :             }
1:M 30 Mar 2026 07:57:25.851 * <timeseries>     { ts-num-threads         :           3 }
1:M 30 Mar 2026 07:57:25.851 * <timeseries>     { ts-retention-policy    :           0 }
1:M 30 Mar 2026 07:57:25.851 * <timeseries>     { ts-duplicate-policy    :       block }
1:M 30 Mar 2026 07:57:25.851 * <timeseries>     { ts-chunk-size-bytes    :        4096 }
1:M 30 Mar 2026 07:57:25.851 * <timeseries>     { ts-encoding            :  compressed }
1:M 30 Mar 2026 07:57:25.851 * <timeseries>     { ts-ignore-max-time-diff:           0 }
1:M 30 Mar 2026 07:57:25.851 * <timeseries>     { ts-ignore-max-val-diff :    0.000000 }
1:M 30 Mar 2026 07:57:25.851 * <timeseries> ]
1:M 30 Mar 2026 07:57:25.851 * <timeseries> Detected redis oss
1:M 30 Mar 2026 07:57:25.851 * <timeseries> Subscribe to ASM events
1:M 30 Mar 2026 07:57:25.851 * <timeseries> Enabled diskless replication
1:M 30 Mar 2026 07:57:25.851 * Module 'timeseries' loaded from /usr/local/lib/redis/modules//redistimeseries.so
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Created new data type 'ReJSON-RL'
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> version: 80600 git sha: unknown branch: unknown
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Exported RedisJSON_V1 API
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Exported RedisJSON_V2 API
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Exported RedisJSON_V3 API
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Exported RedisJSON_V4 API
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Exported RedisJSON_V5 API
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Exported RedisJSON_V6 API
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Enabled diskless replication
```

Name: Gaurav Joshi **Batch: T13** **Roll No:50**

```
1:M 30 Mar 2026 07:57:25.851 * <timeseries> ]
1:M 30 Mar 2026 07:57:25.851 * <timeseries> Detected redis oss
1:M 30 Mar 2026 07:57:25.851 * <timeseries> Subscribe to ASM events
1:M 30 Mar 2026 07:57:25.851 * <timeseries> Enabled diskless replication
1:M 30 Mar 2026 07:57:25.851 * Module 'timeseries' loaded from /usr/local/lib/redis/modules//redistimeseries.so
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Created new data type 'ReJSON-RL'
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> version: 80600 git sha: unknown branch: unknown
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Exported RedisJSON_V1 API
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Exported RedisJSON_V2 API
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Exported RedisJSON_V3 API
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Exported RedisJSON_V4 API
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Exported RedisJSON_V5 API
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Exported RedisJSON_V6 API
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Enabled diskless replication
1:M 30 Mar 2026 07:57:25.851 * <ReJSON> Initialized shared string cache, thread safe: true.
1:M 30 Mar 2026 07:57:25.851 * Module 'ReJSON' loaded from /usr/local/lib/redis/modules//rejson.so
1:M 30 Mar 2026 07:57:25.851 * <search> Acquired RedisJSON_V6 API
1:M 30 Mar 2026 07:57:25.851 * Server initialized
1:M 30 Mar 2026 07:57:25.851 * Ready to accept connections tcp
1:M 30 Mar 2026 07:57:25.851 # WARNING: Redis does not require authentication and is not protected by network restrictions. Redis will accept connections from any IP address on any network interface.
1:signal-handler (1774857755) Received SIGINT scheduling shutdown...
1:M 30 Mar 2026 08:02:35.407 * User requested shutdown...
1:M 30 Mar 2026 08:02:35.407 * Saving the final RDB snapshot before exiting.
1:M 30 Mar 2026 08:02:35.417 * BGSAVE done, 0 keys saved, 0 keys skipped, 88 bytes written.
1:M 30 Mar 2026 08:02:35.422 * DB saved on disk
1:M 30 Mar 2026 08:02:35.422 # Redis is now ready to exit, bye bye...

C:\Users\L15\Desktop\sepm_t13_49>
C:\Users\L15\Desktop\sepm_t13_49>docker ps -a
CONTAINER ID   IMAGE         COMMAND                  CREATED                 STATUS                     PORTS     NAMES
2ced95d05ddb   redis         "docker-entrypoint.s…"   Less than a second ago   Exited (0) 17 seconds ago             nifty_stonebraker
222596f84fc0   hello-world   "/hello"                 3 weeks ago             Exited (0) 3 weeks ago                gracious_wilson

C:\Users\L15\Desktop\sepm_t13_49>docker start nifty_stonebraker
nifty_stonebraker

C:\Users\L15\Desktop\sepm_t13_49>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED                 STATUS                  PORTS      NAMES
2ced95d05ddb   redis     "docker-entrypoint.s…"   Less than a second ago   Up Less than a second   6379/tcp   nifty_stonebraker

C:\Users\L15\Desktop\sepm_t13_49>docker logs nifty_stonebraker
Starting Redis Server
1:C 30 Mar 2026 07:57:25.847 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 30 Mar 2026 07:57:25.847 * Redis version=8.6.2, bits=64, commit=00000000, modified=1, pid=1, just started
1:C 30 Mar 2026 07:57:25.847 * Configuration loaded
1:M 30 Mar 2026 07:57:25.848 * monotonic clock: POSIX clock_gettime
1:M 30 Mar 2026 07:57:25.848 * Running mode=standalone, port=6379.
1:M 30 Mar 2026 07:57:25.848 * <bf> RedisBloom version 8.6.0 (Git=unknown)
1:M 30 Mar 2026 07:57:25.848 * <bf> Registering configuration options: [
1:M 30 Mar 2026 07:57:25.848 * <bf>     { bf-error-rate       :      0.01 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { bf-initial-size     :       100 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { bf-expansion-factor :         2 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-bucket-size      :         2 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-initial-size     :      1024 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-max-iterations   :        20 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-expansion-factor :         1 }
```

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\L15\Desktop\sepm_t13_49>docker --version
Docker version 29.2.1, build a5c7197

C:\Users\L15\Desktop\sepm_t13_49>docker run redis
Unable to find image 'redis:latest' locally
latest: Pulling from library/redis
3e6b2202a764: Pull complete
3f63903b0cb8: Pull complete
f4f2f7018ed9: Pull complete
ec781dee3f47: Pull complete
5f7274725e4f: Pull complete
4f4fb700ef54: Pull complete
c9ff57cee690: Pull complete
3b8772a439dd: Download complete
a4527d030dba: Download complete
Digest: sha256:009cc37796fbdbe1b631b4cc0582bed167e5e403ed8bcd06f77eb6cb5aeb6f93
Status: Downloaded newer image for redis:latest
Starting Redis Server
1:C 30 Mar 2026 07:57:25.847 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 30 Mar 2026 07:57:25.847 * Redis version=8.6.2, bits=64, commit=00000000, modified=1, pid=1, just started
1:C 30 Mar 2026 07:57:25.847 * Configuration loaded
1:M 30 Mar 2026 07:57:25.848 * monotonic clock: POSIX clock_gettime
1:M 30 Mar 2026 07:57:25.848 * Running mode=standalone, port=6379.
1:M 30 Mar 2026 07:57:25.848 * <bf> RedisBloom version 8.6.0 (Git=unknown)
1:M 30 Mar 2026 07:57:25.848 * <bf> Registering configuration options: [
1:M 30 Mar 2026 07:57:25.848 * <bf>     { bf-error-rate       :      0.01 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { bf-initial-size     :       100 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { bf-expansion-factor :         2 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-bucket-size      :         2 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-initial-size     :      1024 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-max-iterations   :        20 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-expansion-factor :         1 }
1:M 30 Mar 2026 07:57:25.848 * <bf>     { cf-max-expansions   :        32 }
1:M 30 Mar 2026 07:57:25.848 * <bf> ]
1:M 30 Mar 2026 07:57:25.849 * Module 'bf' loaded from /usr/local/lib/redis/modules//redisbloom.so
1:M 30 Mar 2026 07:57:25.850 * <search> Redis version found by RedisSearch : 8.6.2 - oss
1:M 30 Mar 2026 07:57:25.850 * <search> RediSearch version 8.6.0 (Git=7782b97)
1:M 30 Mar 2026 07:57:25.850 * <search> Low level api version 1 initialized successfully
1:M 30 Mar 2026 07:57:25.850 * <search> gc: ON, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200, query timeout (ms): 500, timeout policy: return, oom policy: return, cursor read si
ze: 1000, cursor max idle (ms): 300000, max doctable size: 1000000, max number of search results:  1000000, default scorer: BM25STD,
1:M 30 Mar 2026 07:57:25.850 * <search> Initialized thread pools!
1:M 30 Mar 2026 07:57:25.850 * <search> Disabled workers threadpool of size 0
1:M 30 Mar 2026 07:57:25.850 * <search> Subscribe to config changes
1:M 30 Mar 2026 07:57:25.850 * <search> Subscribe to cluster slot migration events
1:M 30 Mar 2026 07:57:25.850 * <search> Enabled role change notification
1:M 30 Mar 2026 07:57:25.850 * <search> Cluster configuration: AUTO partitions, type: 0, coordinator timeout: 0ms
1:M 30 Mar 2026 07:57:25.850 * Module 'search' loaded from /usr/local/lib/redis/modules//redisearch.so
1:M 30 Mar 2026 07:57:25.850 * <timeseries> RedisTimeSeries version 80600, git_sha=05fd355db748676861dc4c17d19c8c1ca74c0154
1:M 30 Mar 2026 07:57:25.851 * <timeseries> Redis version found by RedisTimeSeries : 8.6.2 - oss
1:M 30 Mar 2026 07:57:25.851 * <timeseries> Registering configuration options: [
1:M 30 Mar 2026 07:57:25.851 * <timeseries>     { ts-compaction-policy   :             }
1:M 30 Mar 2026 07:57:25.851 * <timeseries>     { ts-num-threads         :           3 }
1:M 30 Mar 2026 07:57:25.851 * <timeseries>     { ts-retention-policy    :           0 }
```

Name: Gaurav Joshi **Batch: T13** **Roll No:50**

```
1:M 30 Mar 2026 08:21:43.313 * <search> RediSearch version 8.6.0 (Git=7782b97)
1:M 30 Mar 2026 08:21:43.313 * <search> Low level api version 1 initialized successfully
1:M 30 Mar 2026 08:21:43.313 * <search> gc: ON, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200, query timeout (ms): 500, timeout policy: return, oom policy: return, cursor read si
ze: 1000, cursor max idle (ms): 300000, max doctable size: 1000000, max number of search results:  1000000, default scorer: BM25STD,
1:M 30 Mar 2026 08:21:43.313 * <search> Initialized thread pools!
1:M 30 Mar 2026 08:21:43.313 * <search> Disabled workers threadpool of size 0
1:M 30 Mar 2026 08:21:43.313 * <search> Subscribe to config changes
1:M 30 Mar 2026 08:21:43.313 * <search> Subscribe to cluster slot migration events
1:M 30 Mar 2026 08:21:43.313 * <search> Enabled role change notification
1:M 30 Mar 2026 08:21:43.313 * <search> Cluster configuration: AUTO partitions, type: 0, coordinator timeout: 0ms
1:M 30 Mar 2026 08:21:43.313 * Module 'search' loaded from /usr/local/lib/redis/modules//redisearch.so
1:M 30 Mar 2026 08:21:43.313 * <timeseries> RedisTimeSeries version 80600, git_sha=05fd355db748676861dc4c17d19c8c1ca74c0154
1:M 30 Mar 2026 08:21:43.313 * <timeseries> Redis version found by RedisTimeSeries : 8.6.2 - oss
1:M 30 Mar 2026 08:21:43.313 * <timeseries> Registering configuration options: [
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-compaction-policy   :             }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-num-threads         :           3 }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-retention-policy    :           0 }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-duplicate-policy    :       block }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-chunk-size-bytes    :        4096 }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-encoding            :   compressed }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-ignore-max-time-diff:           0 }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-ignore-max-val-diff :    0.000000 }
1:M 30 Mar 2026 08:21:43.314 * <timeseries> ]
1:M 30 Mar 2026 08:21:43.314 * <timeseries> Detected redis oss
1:M 30 Mar 2026 08:21:43.314 * <timeseries> Subscribe to ASM events
1:M 30 Mar 2026 08:21:43.314 * <timeseries> Enabled diskless replication
1:M 30 Mar 2026 08:21:43.314 * Module 'timeseries' loaded from /usr/local/lib/redis/modules//redistimeseries.so
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> Created new data type 'ReJSON-RL'
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> version: 80600 git sha: unknown branch: unknown
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> Exported RedisJSON_V1 API
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> Exported RedisJSON_V2 API
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> Exported RedisJSON_V3 API
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> Exported RedisJSON_V4 API
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> Exported RedisJSON_V5 API
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> Exported RedisJSON_V6 API
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> Enabled diskless replication
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> Initialized shared string cache, thread safe: true.
1:M 30 Mar 2026 08:21:43.314 * Module 'ReJSON' loaded from /usr/local/lib/redis/modules//rejson.so
1:M 30 Mar 2026 08:21:43.314 * <search> Acquired RedisJSON_V6 API
1:M 30 Mar 2026 08:21:43.314 * Server initialized
1:M 30 Mar 2026 08:21:43.314 * Ready to accept connections tcp
1:M 30 Mar 2026 08:21:43.314 # WARNING: Redis does not require authentication and is not protected by network restrictions. Redis will accept connections from any IP address on any network interface.
1:signal-handler (1774858910) Received SIGINT scheduling shutdown...
1:M 30 Mar 2026 08:21:50.138 * User requested shutdown...
1:M 30 Mar 2026 08:21:50.138 * Saving the final RDB snapshot before exiting.
1:M 30 Mar 2026 08:21:50.139 * BGSAVE done, 0 keys saved, 0 keys skipped, 88 bytes written.
1:M 30 Mar 2026 08:21:50.154 * DB saved on disk
1:M 30 Mar 2026 08:21:50.154 # Redis is now ready to exit, bye bye...

C:\Users\L15\Desktop\sepm_t13_49>
```

```
hello-world:latest          ef54e839ef54       25.9kB         9.52kB    U
myredis:version1            9199aba240ab        202MB         53.2MB
redis:latest                009cc37796fb        204MB         55.3MB    U

C:\Users\L15\Desktop\sepm_t13_49>docker run redis
Starting Redis Server
1:C 30 Mar 2026 08:21:43.310 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 30 Mar 2026 08:21:43.310 * Redis version=8.6.2, bits=64, commit=00000000, modified=1, pid=1, just started
1:C 30 Mar 2026 08:21:43.310 * Configuration loaded
1:M 30 Mar 2026 08:21:43.311 * monotonic clock: POSIX clock_gettime
1:M 30 Mar 2026 08:21:43.311 * Running mode=standalone, port=6379.
1:M 30 Mar 2026 08:21:43.311 * <bf> RedisBloom version 8.6.0 (Git=unknown)
1:M 30 Mar 2026 08:21:43.311 * <bf> Registering configuration options: [
1:M 30 Mar 2026 08:21:43.311 * <bf>     { bf-error-rate       :      0.01 }
1:M 30 Mar 2026 08:21:43.311 * <bf>     { bf-initial-size     :       100 }
1:M 30 Mar 2026 08:21:43.311 * <bf>     { bf-expansion-factor :         2 }
1:M 30 Mar 2026 08:21:43.311 * <bf>     { cf-bucket-size      :         2 }
1:M 30 Mar 2026 08:21:43.311 * <bf>     { cf-initial-size     :      1024 }
1:M 30 Mar 2026 08:21:43.311 * <bf>     { cf-max-iterations   :        20 }
1:M 30 Mar 2026 08:21:43.311 * <bf>     { cf-expansion-factor :         1 }
1:M 30 Mar 2026 08:21:43.311 * <bf>     { cf-max-expansions   :        32 }
1:M 30 Mar 2026 08:21:43.311 * <bf> ]
1:M 30 Mar 2026 08:21:43.312 * Module 'bf' loaded from /usr/local/lib/redis/modules//redisbloom.so
1:M 30 Mar 2026 08:21:43.313 * <search> Redis version found by RedisSearch : 8.6.2 - oss
1:M 30 Mar 2026 08:21:43.313 * <search> RediSearch version 8.6.0 (Git=7782b97)
1:M 30 Mar 2026 08:21:43.313 * <search> Low level api version 1 initialized successfully
1:M 30 Mar 2026 08:21:43.313 * <search> gc: ON, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200, query timeout (ms): 500, timeout policy: return, oom policy: return, cursor read si
ze: 1000, cursor max idle (ms): 300000, max doctable size: 1000000, max number of search results:  1000000, default scorer: BM25STD,
1:M 30 Mar 2026 08:21:43.313 * <search> Initialized thread pools!
1:M 30 Mar 2026 08:21:43.313 * <search> Disabled workers threadpool of size 0
1:M 30 Mar 2026 08:21:43.313 * <search> Subscribe to config changes
1:M 30 Mar 2026 08:21:43.313 * <search> Subscribe to cluster slot migration events
1:M 30 Mar 2026 08:21:43.313 * <search> Enabled role change notification
1:M 30 Mar 2026 08:21:43.313 * <search> Cluster configuration: AUTO partitions, type: 0, coordinator timeout: 0ms
1:M 30 Mar 2026 08:21:43.313 * Module 'search' loaded from /usr/local/lib/redis/modules//redisearch.so
1:M 30 Mar 2026 08:21:43.313 * <timeseries> RedisTimeSeries version 80600, git_sha=05fd355db748676861dc4c17d19c8c1ca74c0154
1:M 30 Mar 2026 08:21:43.313 * <timeseries> Redis version found by RedisTimeSeries : 8.6.2 - oss
1:M 30 Mar 2026 08:21:43.313 * <timeseries> Registering configuration options: [
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-compaction-policy   :             }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-num-threads         :           3 }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-retention-policy    :           0 }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-duplicate-policy    :       block }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-chunk-size-bytes    :        4096 }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-encoding            :   compressed }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-ignore-max-time-diff:           0 }
1:M 30 Mar 2026 08:21:43.314 * <timeseries>     { ts-ignore-max-val-diff :    0.000000 }
1:M 30 Mar 2026 08:21:43.314 * <timeseries> ]
1:M 30 Mar 2026 08:21:43.314 * <timeseries> Detected redis oss
1:M 30 Mar 2026 08:21:43.314 * <timeseries> Subscribe to ASM events
1:M 30 Mar 2026 08:21:43.314 * <timeseries> Enabled diskless replication
1:M 30 Mar 2026 08:21:43.314 * Module 'timeseries' loaded from /usr/local/lib/redis/modules//redistimeseries.so
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> Created new data type 'ReJSON-RL'
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> version: 80600 git sha: unknown branch: unknown
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> Exported RedisJSON_V1 API
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> Exported RedisJSON_V2 API
1:M 30 Mar 2026 08:21:43.314 * <ReJSON> Exported RedisJSON_V3 API
```

Conclusion:

Docker revolutionizes the software development and deployment process by providing a powerful platform for containerization. By encapsulating applications and their dependencies into lightweight, portable containers,

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